

Fungus inspired semiconductors for flexible electronics

Researchers at Pennsylvania State University (Penn State), University Park, USA, have developed a bio-hybrid memory



From fungi to flexible electronics: Advancing nature-inspired semiconductor solutions (Credit: <https://www.psu.edu>)

device by integrating synthetic DNA with perovskite semiconductors.

The system forms a low-power memristor capable of storing and processing data simultaneously, consuming up to 100 times less energy than conventional storage. The breakthrough could

enable high-capacity, energy-efficient data storage solutions for AI and next-generation computing systems.

Catalyst that could change fuel cells forever

A newly developed catalyst could significantly improve fuel cell efficiency and durability, potentially reducing reliance on expensive materials like platinum. The innovation enhances key electrochemical reactions, enabling more stable and cost-effective energy conversion in hydrogen fuel cells.



Nickel catalyst enables high-performance fuel cell free of precious metals (Credit: <https://news.cornell.edu>)

Engineered by researchers at Cornell University, United States, the catalyst

uses an organic, carbon-based molecule that remains stable under operating conditions while maintaining high activity. This approach addresses a major limitation of conventional catalysts, which often degrade over time. The breakthrough could accelerate the adoption of fuel cells in electric vehicles and clean energy systems, supporting the shift toward sustainable, low-emission power technologies.

Humanoid robotic hand enables delicate real-world tasks

A new humanoid robotic hand can now sense its own finger positions in real time, enabling far more precise and human-like manipulation of objects. The system provides continuous feedback on finger bending and side-to-side movement, improving control during complex grasping tasks. Developed by researchers at Zhejiang University,

along with Hangzhou Dianzi University and Lishui University, the hand integrates an omnidirectional soft-bending sensor into each finger. This allows it to track multi-degree-of-freedom motion using optical sensing and light-based signal analysis inside flexible fibre structures. The advancement could improve humanoid robotics, prosthetics, and precision automation by giving machines a more accurate sense of self-movement, enabling delicate real-world tasks such as tool use and object handling.

Low-cost, nextgen electronics based on electron motion

Researchers at North Carolina State University, USA, have developed a new method to generate orbital current using



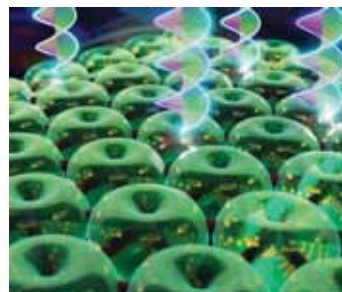
Chiral Phonons (Credit: <https://news.ncsu.edu>)

chiral phonons, atomic vibrations that transfer angular momentum directly to electrons. The approach eliminates the need for magnets, volt-

age, or rare materials, enabling simpler and energy-efficient orbitronic devices. The breakthrough could pave the way for low-cost, next-generation electronics based on electron orbital motion.

Optical tornadoes enable next-gen lasers breakthrough

A breakthrough in photonics has enabled 'optical tornadoes', spiralling light structures, that could lead to more stable and efficient next-generation lasers. The technique



A tiny optical trap lets light twist and persist in its lowest-energy state, opening new possibilities for scalable photonic technologies (Credit: <https://scitechdaily.com>)

allows light to twist and persist in a controlled, low-energy state, improving coherence and scalability for laser systems. Developed by researchers at the University of Warsaw, the system uses liquid crystals to form microscopic structures that trap and rotate light using synthetic magnetic fields. An optical micro-cavity further stabilises

the effect, while external voltage enables precise control. The innovation could advance compact lasers, optical communication, and quantum technologies, offering a simpler, scalable alternative to complex photonic designs.